IMEN JARRAYA

Postdoctoral Fellow RIOTU Lab

CONTACT

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+966 55 725 44 96

Location

Riyadh, Saudi Arabia

EDUCATION

Ph.D. Electrical Eng.

ENIS & ESTACA 2016–2021

M.Sc. Electrical Conversion

ENIS, 2008–2010

SKILLS

Simulation

Matlab, Simulink, LabView

Programming

C/C++, Python, Android

Robotics

ROS 1 & 2, Gazebo, PX4

Hardware

Arduino, STM32, Jetson

LANGUAGES

Arabic Native

French Fluent

English Professional

AWARDS

Women in Science, AIDA Lab, PSU 2025

Special Session Chair, ICCSIC 2026

IMEN JARRAYA

Electrical Engineer · Postdoctoral Fellow

PROFESSIONAL SUMMARY

Ph.D. Electrical Engineer and Postdoctoral Fellow at Prince Sultan University (RIOTU Lab, Riyadh), combining academic research, university teaching, and final-year project supervision. Research focuses on autonomous UAV navigation in GNSS-denied environments, hybrid EV energy management, embedded systems, and IoT/AI integration. Author of multiple Q1 journal articles (IF up to 10.1), 11 conference papers, and 4 grant proposals as Principal Investigator totaling approximately 6 million SAR.

PROFESSIONAL EXPERIENCE

Oct 2023 – Present • Postdoctoral Fellow

Prince Sultan University (PSU), RIOTU Lab · Riyadh

Leading high-impact Q1 journal publications in UAV navigation and embedded systems. Submitted 4 competitive research grant proposals as Principal Investigator (~6M SAR). Supervised research projects and fostered interdisciplinary academic collaboration.

2021 – 2023 • University Teaching Assistant

Higher Institute of Technological Studies (ISET) · Sidi Bouzid, Tunisia

Delivered lectures and laboratory sessions in Real-Time Systems, Embedded Systems Programming, and IoT. Designed course materials and supervised end-of-studies projects.

2010 – 2015 • University Lecturer

ISGIS & Institute of Biotechnology · Sfax, Tunisia

Taught Industrial Computing, Power Electronics, Automation, Electrical Engineering, and Biomedical Measurement. Designed laboratory exercises and mentored students.

2020 – 2021 • Research & Development Engineer

Novel-Ti · Tunisia

Led R&D; in embedded systems, robotics, IoT, and artificial intelligence. Developed PCBs using Altium Designer and implemented hoverboard tilt control systems.

EDUCATION

Ph.D. in Electrical Engineering · 2016–2021

National School of Engineers of Sfax (ENIS) & ESTACA, France | Thesis: Energy management of hybrid EV with Li-ion battery and supercapacitors | Distinction: Very Honorable

M.Sc. Electrical Conversion & Renewable Energies · 2008–2010

National School of Engineers of Sfax (ENIS) | Thesis: Decision support tool for wind power systems adapted to Tunisian sites | Rank: 5/33

KEY PUBLICATIONS

• CLAK: CNN-LSTM-Attention and KAN for non-visual UAV localization in GNSS-denied environments. **Satellite Navigation** (Q1, IF=10.1) · 2026

• Two-Stage Residual EKF using XGBoost and KAN for terrain-aided UAV localization. **Engineering Applications of AI** (Q1, IF=8.0) · 2026

• SOH-KLSTM: Hybrid Kolmogorov-Arnold Network and LSTM for Li-ion battery health monitoring. **Journal of Energy Storage** (Q1, IF=9.8) · 2025